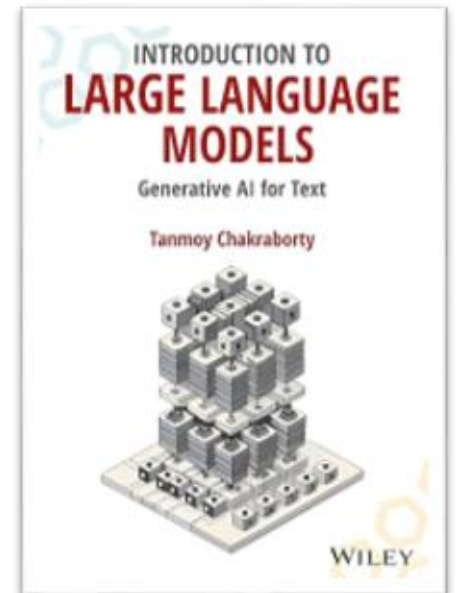


Neural Language Models



Tanmoy Chakraborty
Associate Professor, IIT Delhi
<https://tanmoychak.com/>



PC

unigram

(LMI)

Accuracy

$$P(w_1, w_2, \dots, w_n) = P(w_1)P(w_2) \dots P(w_n)$$

bigram
LM2

$$P(w_1) \times P(w_2|w_1) \times P(w_3|w_2) \dots$$

$$P(w_1, w_2, \dots, w_n) = 1$$



Entropy

Task-based Evaluation / Extrinsic Eval

(LMI)

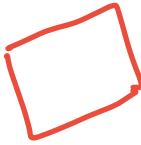
Tagore's Corp

Task

1) Fill in the blanks.

(LM2)

The cat sat on the _____!



Spelling Correction



NT

Intrinsic Eval

Task-agnostic Eval

Perplexity (PP)

$$= P(w_1, w_2, \dots, w_n)^{-1/n}$$

$$= \frac{1}{P(w_1, w_2, \dots, w_n)^{1/n}}$$

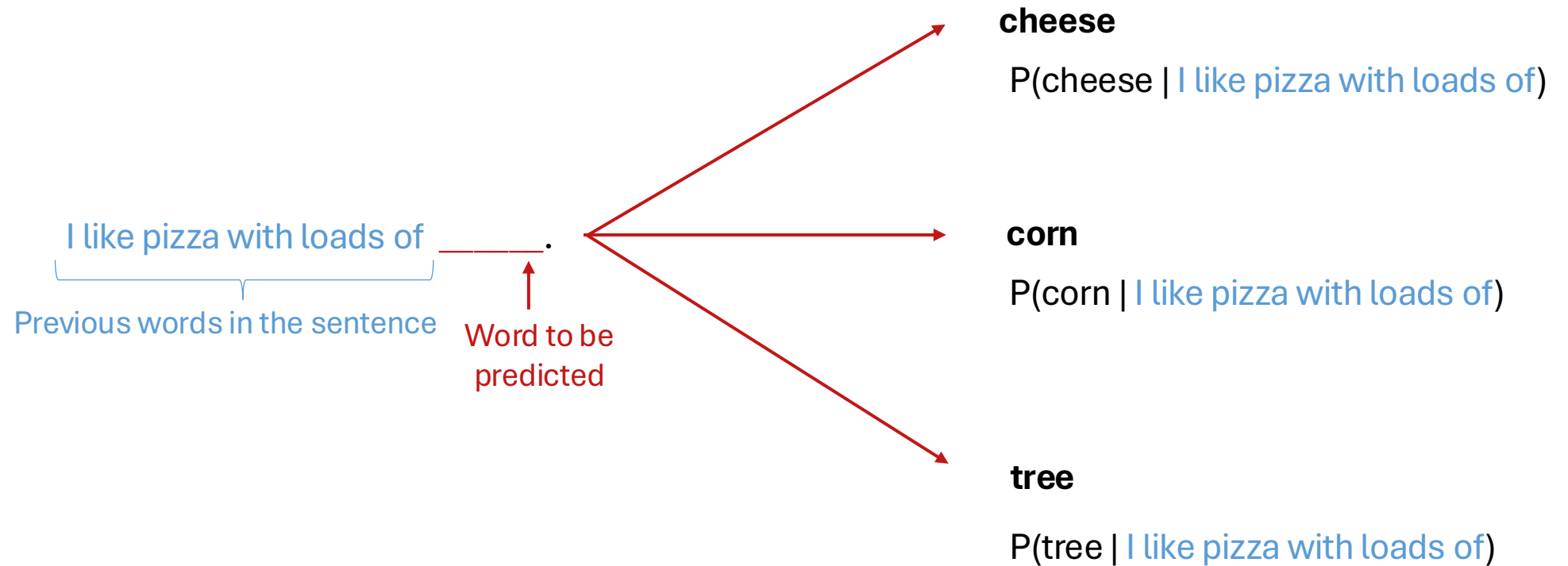
$$= \frac{1}{\sqrt[n]{P(w_1, w_2, \dots, w_n)}}$$

$$LM(1) = \frac{1}{\sqrt[n]{P(w_1)P(w_2) \dots P(w_n)}}$$

$$LM(2) = \frac{1}{\sqrt[n]{P(w_2)P(w_3|w_2) \dots}}$$

Language Modeling

- **Language Modeling** is the task of predicting what word comes next



Language Modeling

- You can also think of a Language Model as a system that assigns a probability to a piece of text.
- For example, if we have some text $x^{(1)}, \dots, x^{(T)}$, then the probability of this text (according to the Language Model) is:

$$\begin{aligned} P(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(T)}) &= P(\mathbf{x}^{(1)}) \times P(\mathbf{x}^{(2)} | \mathbf{x}^{(1)}) \times \dots \times P(\mathbf{x}^{(T)} | \mathbf{x}^{(T-1)}, \dots, \mathbf{x}^{(1)}) \\ &= \prod_{t=1}^T \underbrace{P(\mathbf{x}^{(t)} | \mathbf{x}^{(t-1)}, \dots, \mathbf{x}^{(1)})} \end{aligned}$$

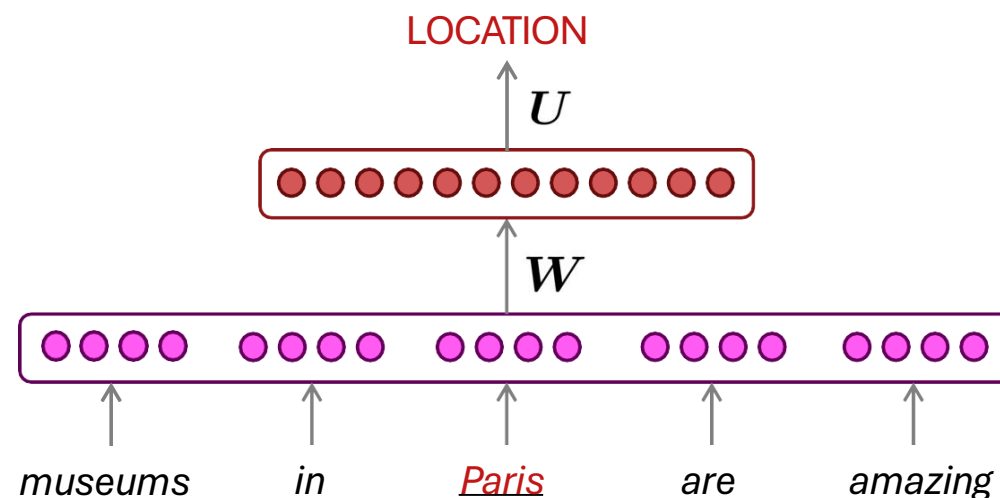
This is what our LM provides



How to Build a *Neural* Language Model?

- Recall the Language Modeling task:
 - Input:** sequence of words $x^{(1)}, x^{(2)}, \dots, x^{(t)}$
 - Output:** probability distribution of the next word $P(x^{(t+1)} | x^{(t)}, \dots, x^{(1)})$
- How about a window-based neural model?

Example: NER Task



A Fixed-window Neural Language Model

~~as the proctor started the clock~~
discard

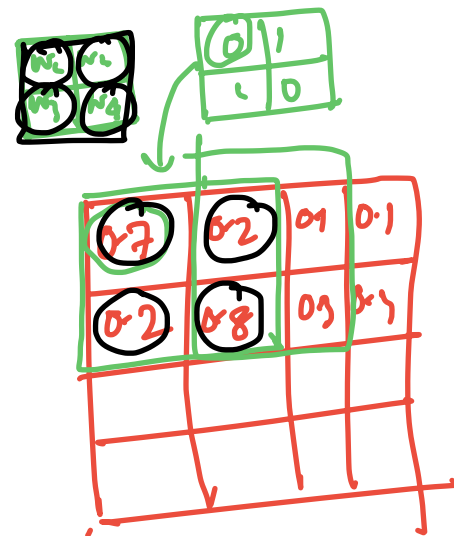
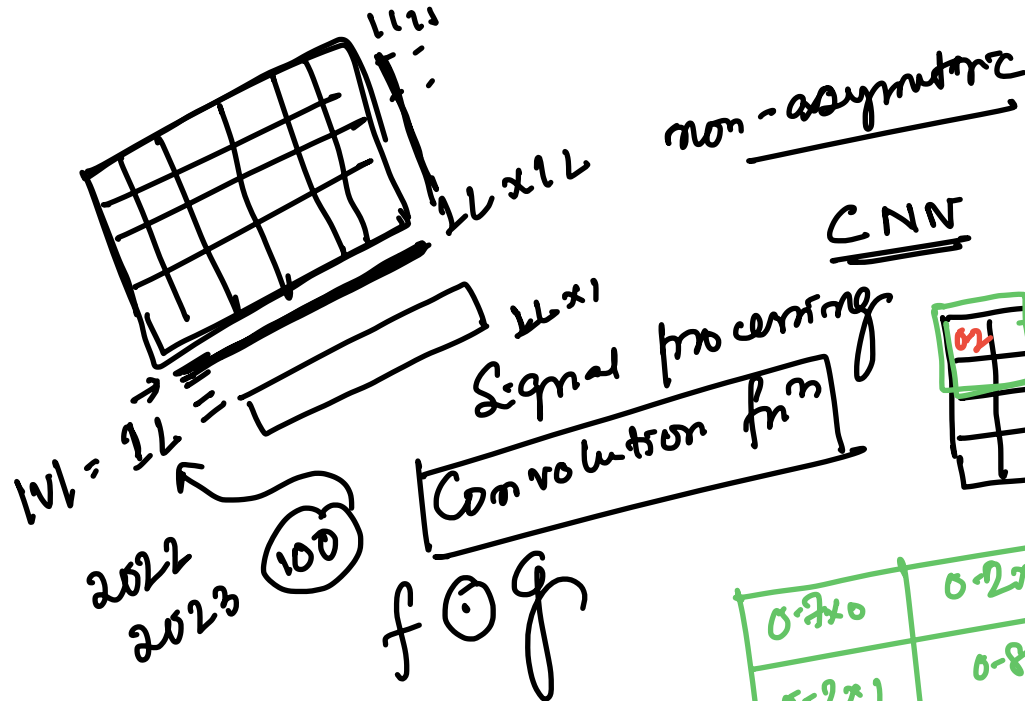
the students opened their _____
fixed window



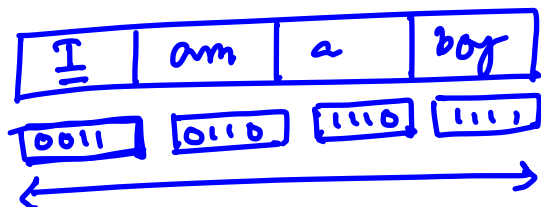
Problem of SLM

1) Sparsity

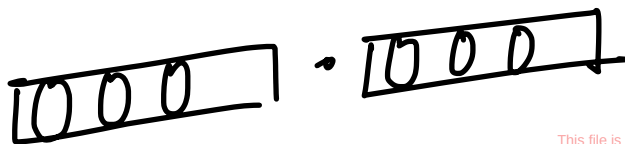
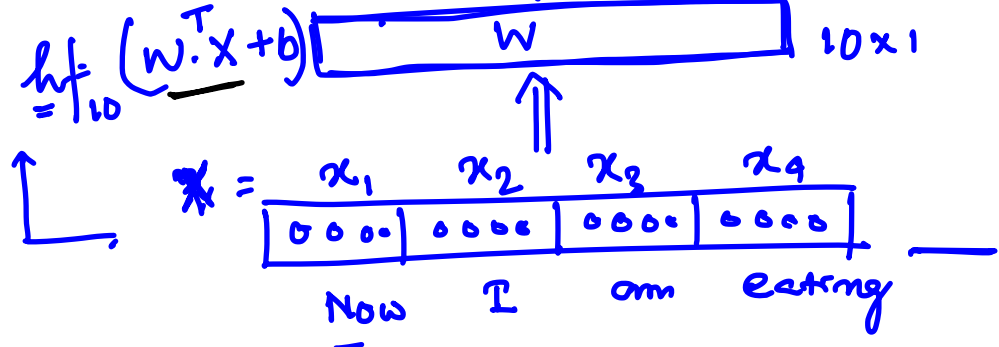
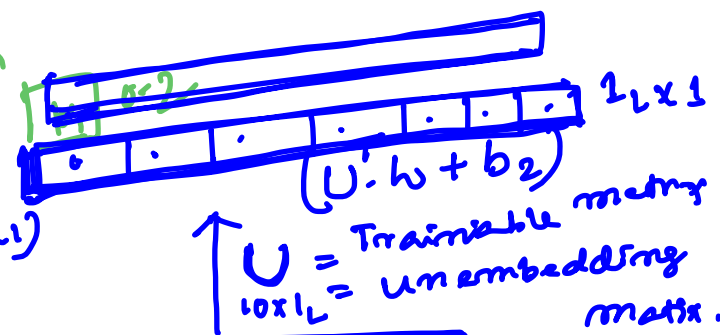
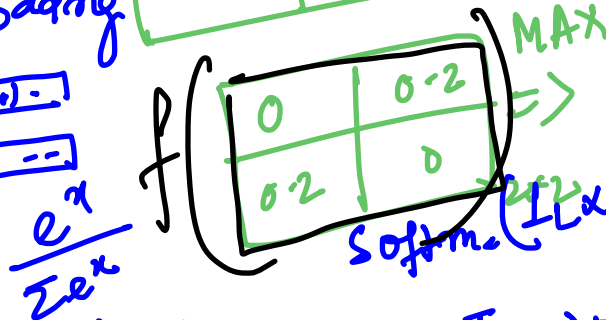
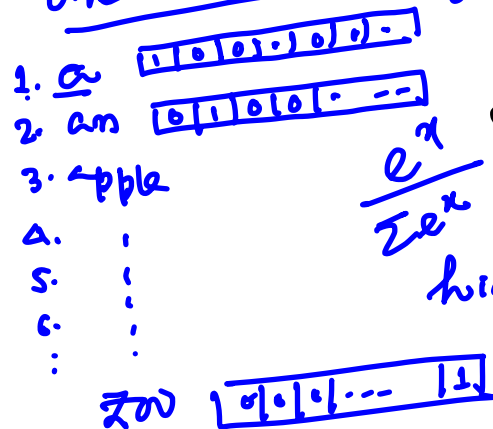
2)

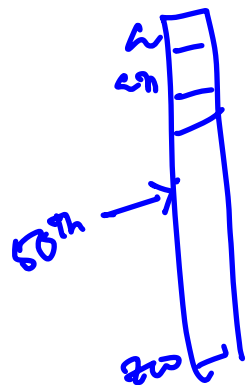


I am a boy

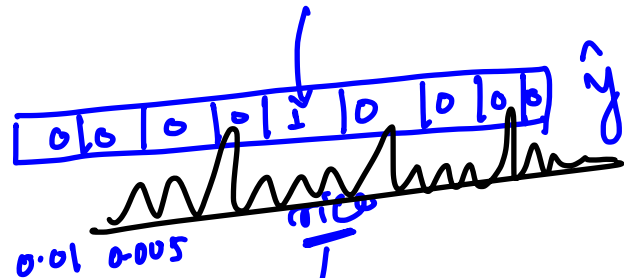


one-hot encoding



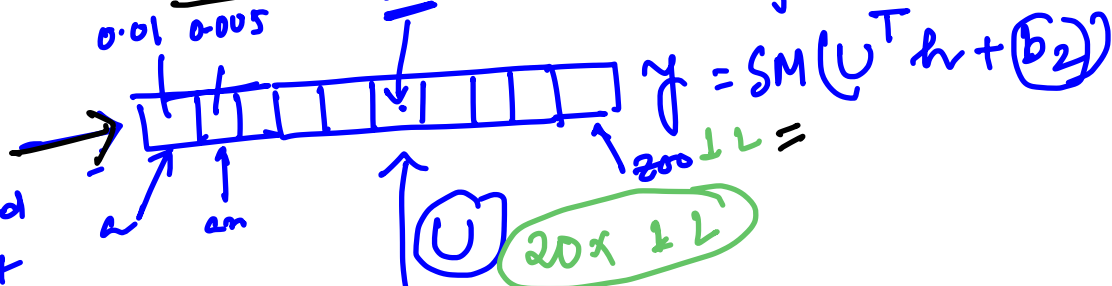


Ground truth prob dist



CE
Cross entropy ($y, \hat{y}$)

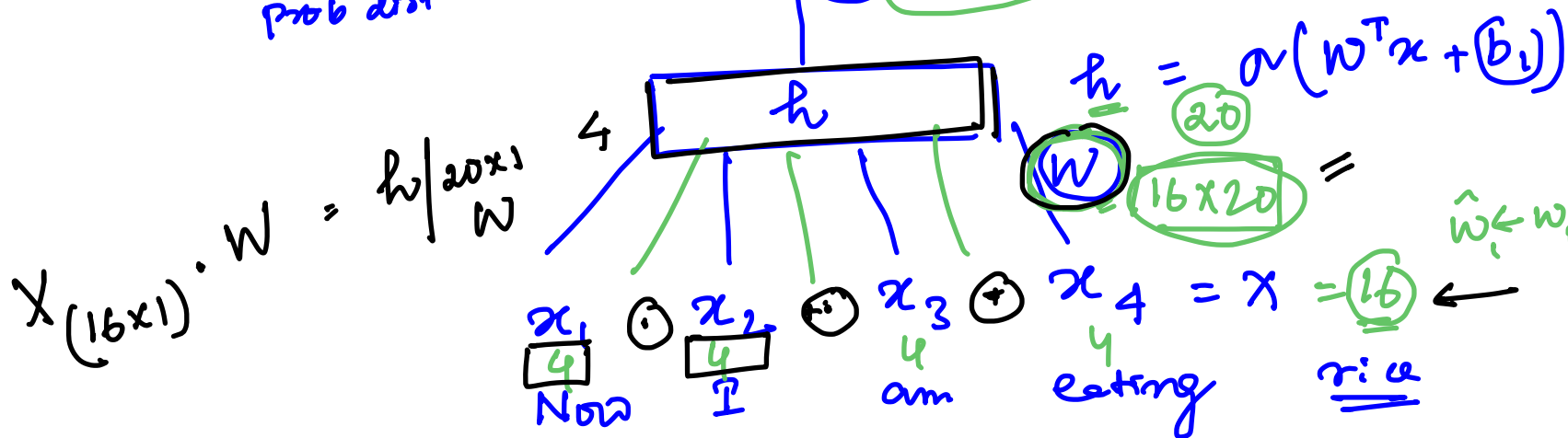
Predicted prob dist



$$y = (y_1, y_2, \dots, y_m)$$

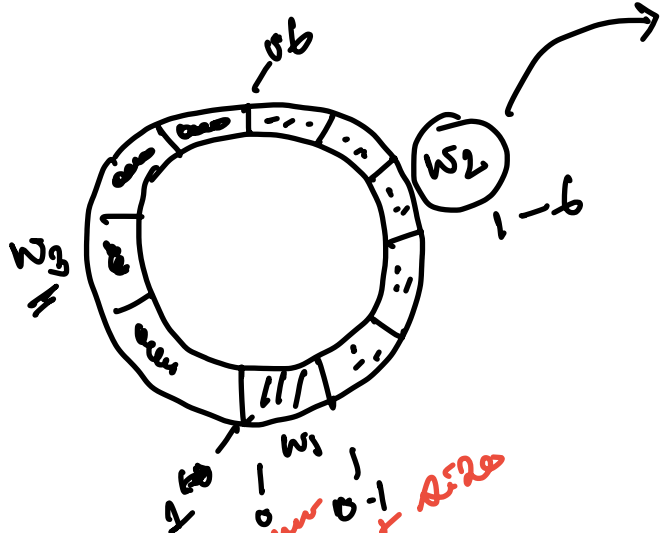
$$\hat{y} = (\hat{y}_1, \hat{y}_2, \dots, \hat{y}_m)$$

$$CE(y, \hat{y}) = -y_1 \log \hat{y}_1 - y_2 \log \hat{y}_2 - y_3 \log \hat{y}_3 - \dots$$

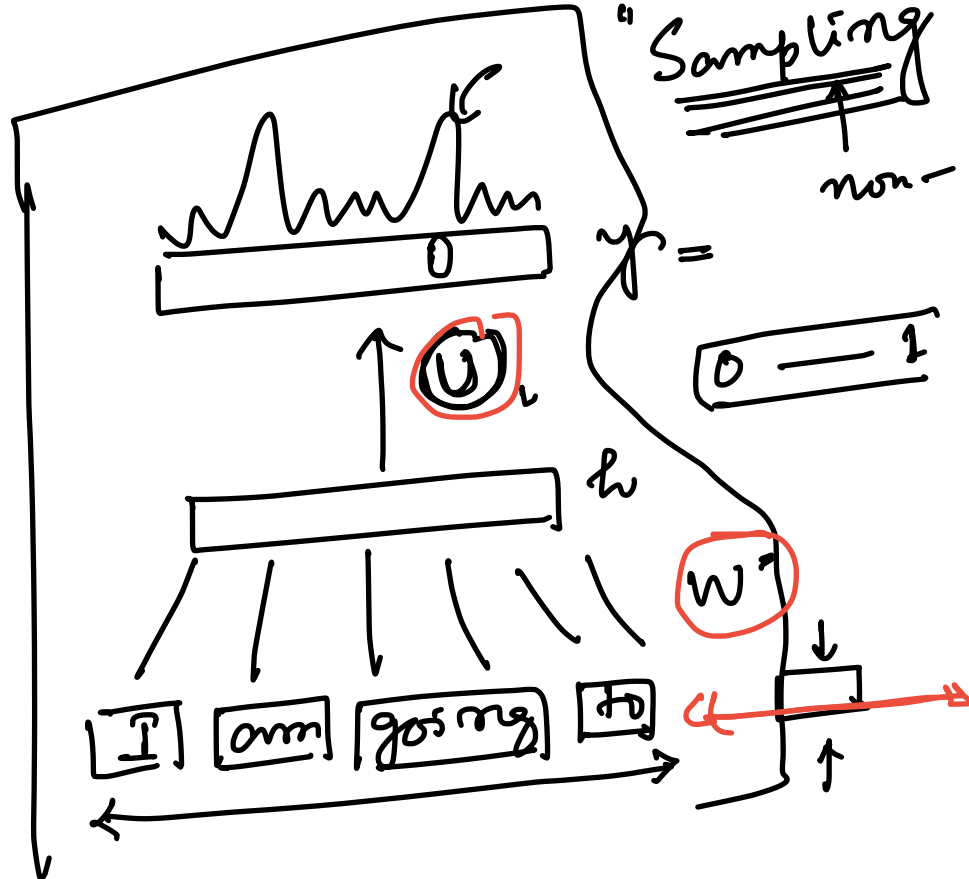


$$\hat{w}_i \leftarrow w_i - \eta \left[\frac{\partial CE}{\partial w_i} \right]$$

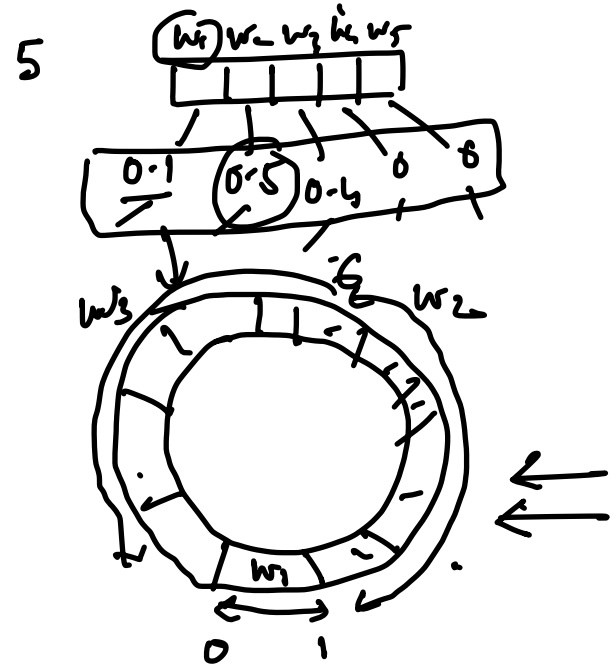
- ① Sparsity prob
- ② no need to store bigram matrix W, U



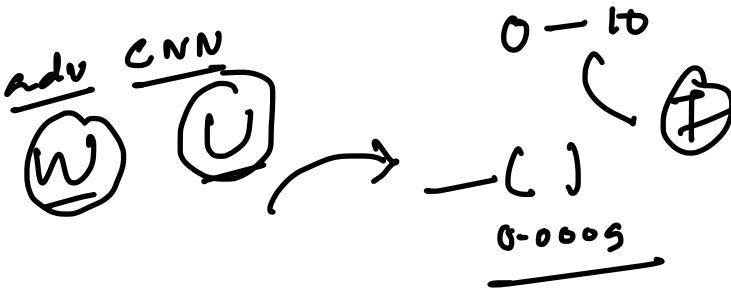
- ① need to determine size
- ② dim of W and U will change if context length and vocab will change

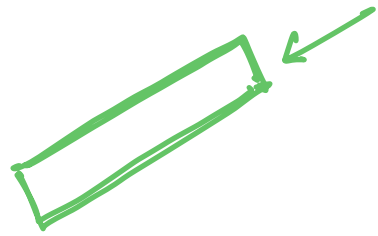


"Sampling" based on prob.
non-deterministic / greedy =



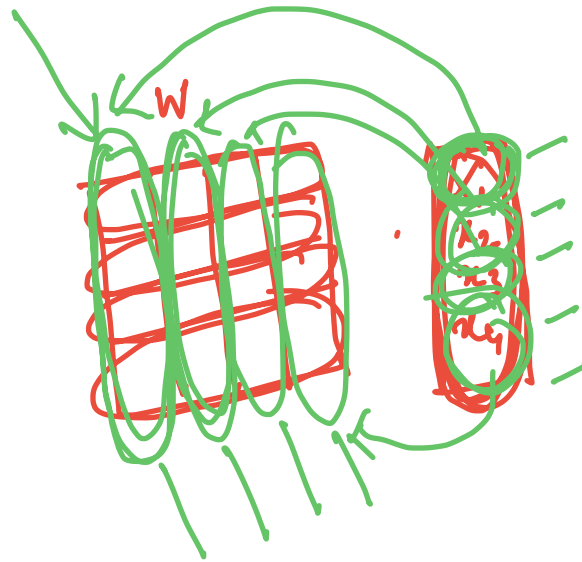
SLM





$\sum_{i=1}^T x_i$
 $\phi(x)$

RNN



- ✓ ① Content length ()
- ✓ ② w, v —
 ↑
- ✓ ③ lack of memory
 padding